PalmSource™ 99

How to Localize Existing Applications

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Agenda

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- Internationalization vs. localization
- Areas of challenge
- Preparing for a localization project
- Localizing Palm OS® applications
- Tips and Techniques
- Resources
- Summary
- Meeting the Palm localization team
- Q&A

Internationalization vs. Localization

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- Internationalization (I18n)
  - Writing applications that will function in more than one language
  - e.g., Generic code base, separating code from localizable resources, etc.
- Localization (L10n)
  - Adapting a product to a particular language or region
  - e.g., Translation, dialog resizing, customizing default data, etc.

Areas of Challenge

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- General formats
  - Numbers:

U.S.	1,234.00
Italy	1.234,00
Sweden	1 234,00

  - Date:

U.S.	February 14, 1999
Sweden	99-14-02
Italy	14.2.99

  - Time:

U.S.	7:23 PM
Germany	19.23
French-Canadian	19 h 23

Areas of Challenge

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- Cultural sensitivity
  - Icons, bitmaps, color
- Same language, different dialect
  - English—U.S. vs. U.K.
  - Spanish—Mexico vs. Spain
- Region-specific requirements
  - Rules for abbreviation
  - Encryption export control restrictions
  - Phone numbers, URLs, Internet service providers, etc.

Preparing for a Localization Project

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- Knowing when to start
- Choose professional and experienced localizers
- Glossary = translation = consistency
- Localization kit

Localizing Palm OS® Applications

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- Setting up multi-lingual development environment

```
project\
    source\
    resources\
        English\
        Japanese\
        French\
        ...
```

Localizing Palm OS® Applications

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- Resources extraction
  - Macintosh tools—AppleGlot, PowerGlot, MPW rez/derez, etc.
  - Windows tools—Trados, Corel Catalyst
- Dialog resizing
  - Macintosh Tools—ResEdit, Resorcerer, Metrowerks Constructor
  - Windows tools—Metrowerks Constructor for Windows

Tips/Techniques

- How to handle special characters
  - ® ---> (R)
  - ™ ---> TM
  - … ---> … (3 periods)







Tips/Techniques

- When to localize with .prc level resource editing tool—OnBoard RsrcEdit, etc.
- Use half-width Japanese characters to maximize space



Resources

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- www.lisa.org:
  - Great starting point for tools, technologies, and information related to localization
- Term glossary
  - Apple's term glossaries:
    ftp://dev.apple.com/devworld/Tool_Chest/Localization_Tools/Apple_Intl_Glossaries/
  - Microsoft's language glossaries:
    ftp://ftp.microsoft.com/developr/msdn/newup/glossary
  - Palm OS® SDK:
    - Japanese term glossary included
    - European term glossaries in the works

Summary

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- Localize with cultural awareness
  - Do not make any locale specific assumptions in your code
- Good preparation does payoff
  - Design your software to be localizable from the beginning
- Learn from our, and other's mistakes
  - Apply the tips and techniques on localization that you've learned from PalmSource


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Q&A

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